

2024



Phillips 66 Resolve Observability Solution

Learn how Phillips 66 in the energy industry uses AWS, Grafana and Amazon Managed Service for Prometheus to improve observability.



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30% reduction

in mean time to resolve issues for refining business

Reduced downtime

with faster resolution and issue prevention

Empowered

Response Center to resolve issues after hours

Increased

employee productivity and independence

Overview

Global energy company [Phillips 66](#) depends on thousands of applications to supply energy to people around the globe. To keep the applications running, Phillips 66 employs 30 DevOps teams to monitor multiple observability solutions. Seeking greater visibility across that volume of data, the company looked to Amazon Web Services (AWS) to centralize and simplify the monitoring of its hybrid and multicloud environments.

Using services such as [Amazon Managed Service for Prometheus](#)—which provides highly available, secure, and managed monitoring for containerized systems—Phillips 66 achieved its observability goals by providing employees with visibility across the high volumes of data that are used and produced by these applications. This single source of truth facilitates data-driven decision-making across Phillips 66, helping the company respond to events with agility and precision.



Opportunity | Using Amazon Managed Service for Prometheus to Launch an Observability Solution for Phillips 66

Based in the United States, Phillips 66 has nearly 150 years of experience in the energy industry. The energy manufacturing and logistics company maintains businesses in refining, midstream, chemicals, and marketing and specialties. The company runs 70 percent of compute using AWS services, but it also maintains a hybrid and multicloud infrastructure.

While the bulk of Phillips 66's workloads run on AWS, the company runs several custom line-of-business applications in on-premises data centers and hosts its enterprise resource planning system on Azure. This technology stack, while expansive, posed several challenges. The company had to use multiple observability tools to manage this massive portfolio. However, using multiple tools led to a fragmented system with limited visibility. Phillips 66 lacked a clear view of critical services, such as its LIMS application, which is essential for maintaining operational quality and compliance. This fragmentation limited the company's ability to resolve issues quickly and effectively. To overcome these challenges, it began to search for a centralized monitoring and observability solution in 2020.

During this time, Phillips 66 tested several proof-of-concept builds from different vendors. AWS emerged as the clear choice for its project; the company found that it could meet its security and data volume requirements on AWS, and it gained access to high-quality support services. Crucially, it could monitor all of its applications on AWS, including applications hosted on premises and with other cloud providers.

Phillips 66 engaged [AWS Professional Services](#), which helps organizations achieve desired business outcomes using AWS, to help design and implement a new monitoring and observability solution. "Meeting with teams from AWS weekly and working closely together gave us a level of trust that AWS wasn't just selling us a product," says Shane Carter, director of cloud operations at Phillips 66. "AWS is there every day to help solve any problems that arise and make sure the products fulfill our needs."

Solution | Reducing Mean Time to Resolve Issues by 30% Using Observability Solution Built on AWS

In 3 months, Phillips 66 built its observability solution with the support of the AWS Professional Services team, who helped the company identify the optimal AWS services to serve as the solution's foundation. '  challenge

came up, AWS Professional Services had deep knowledge or knew another team to reach out to for help,” says Carter.

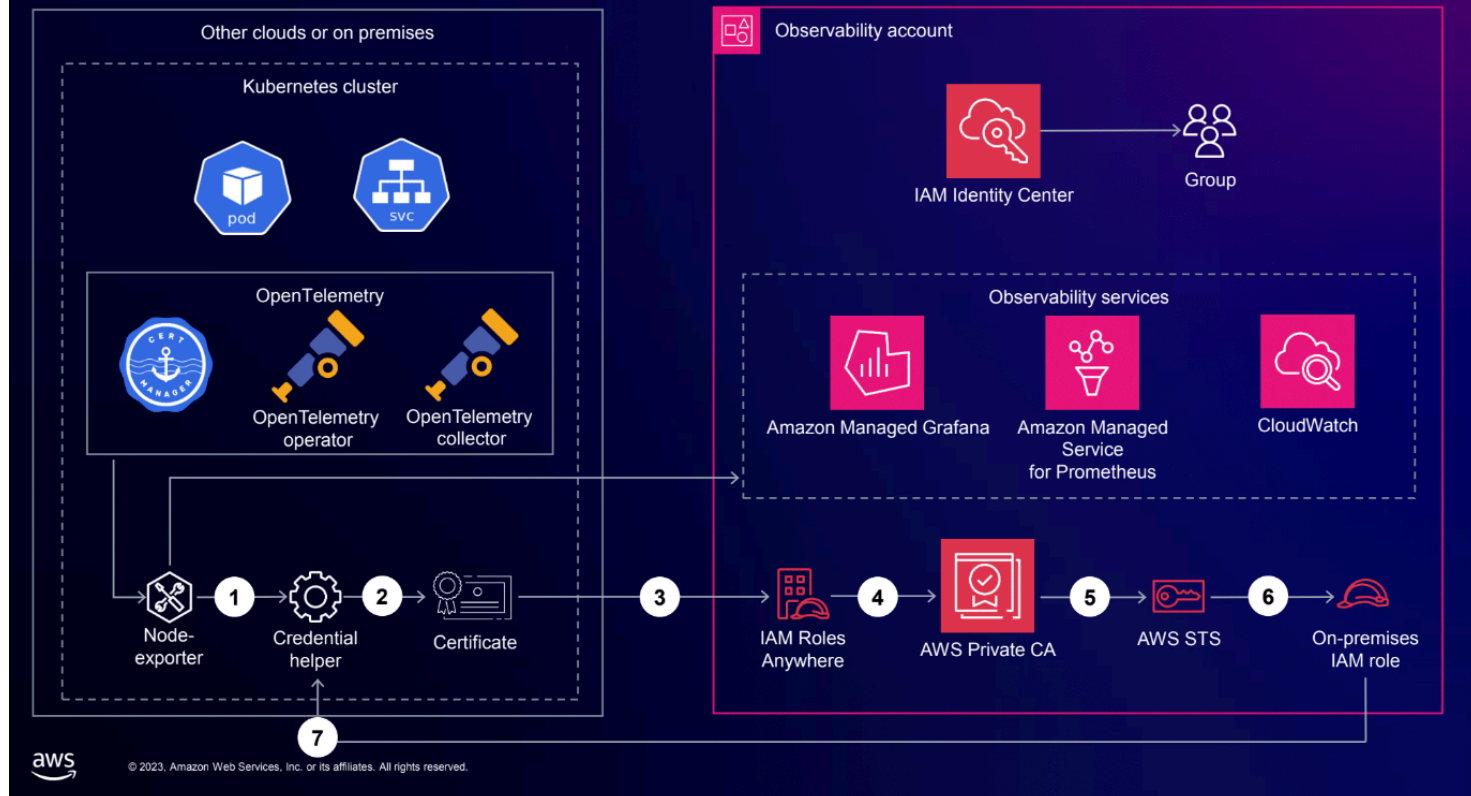
Using Amazon Managed Service for Prometheus, Phillips 66 can gather data from various tools, including [Amazon CloudWatch](#), which observes and monitors resources and applications on AWS, on premises, and on other clouds. Regardless of whether an application runs on AWS, another cloud provider, on premises, or a combination, all data then appears on a single dashboard using [Amazon Managed Grafana](#), a scalable and secure data visualization service for operational metrics, logs, and traces. Phillips 66 can create custom alerts for application performance using Amazon Managed Service for Prometheus and create, explore, and share observability dashboards using Amazon Managed Grafana. This increased visibility means teams can work more efficiently.

Phillip 66’s DevOps team in the refining business estimates that the observability solution built on AWS reduces the mean time to resolve issues by 30 percent. Instead of needing to open multiple tools from different vendors, the team can see all metrics together using Amazon Managed Grafana, making it simpler to spot and resolve issues. Previously, Phillips 66 didn’t have any visibility into its LIMS application, which is used for quality control. Now, the company can monitor trends and set up automation. For example, it can notify the relevant DevOps team to investigate if metrics exceed a threshold, or the company can automatically restart a service before an outage. Phillips 66 reduces downtime because teams can prevent and resolve issues faster using the dashboard and proactive notifications.

Phillips 66 staffs its Response Center 24/7 as the first line of defense for issues. If a critical issue came in after hours, employees at the Response Center often needed to pull in experts to access relevant data. Using AWS, Phillips 66 makes all data available across the company so that the Response Center can resolve issues faster and more independently. “We’re driving better business decisions because we can see the data,” says Carter. “Having a Response Center with the capability of fixing an issue at night or on the weekend without pulling in 20 more people is amazing because it reduces the time to get the application back up and running and frees up time for everybody else.”

With its solution built on AWS, Phillips 66 simplifies and centralizes the observability of its application portfolio and gives its teams full access to their data so that they can be more productive. “It takes my cloud operations team out of the equation and gives power back to the support teams to do their own debugging,” says Carter.

Architecture Diagram



Outcome | Expanding to Use Artificial Intelligence to Prevent Issues Using AWS

Phillips 66 plans to continue investing in its observability solution and incorporating artificial intelligence and machine learning in the future to prevent more issues before they occur and automate fixes to reduce manual intervention. "One of the biggest benefits of our observability solution built on AWS is that we have one place to go," says Carter. "No matter who you are in the company, you can go to one place to access data."

About Phillips 66

Phillips 66 is an energy manufacturing and logistics company with nearly 150 years of experience in the industry. The company has a vision to provide energy and improve lives around the world.

AWS Services Used

Amazon Managed Service for Prometheus

Amazon Managed Service for Prometheus is a Prometheus-compatible service that monitors and provides alerts on containerized applications and infrastructure at scale.

[Learn more »](#)



Amazon Managed Grafana

Amazon Managed Grafana is a fully managed service for Grafana, a popular open-source analytics platform that enables you to query, visualize, and alert on your metrics, logs, and traces.

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Amazon CloudWatch

Amazon CloudWatch is a service that monitors applications, responds to performance changes, optimizes resource use, and provides insights into operational health.

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AWS Professional Services

The AWS Professional Services organization is a global team of experts that can help you realize your desired business outcomes when using the AWS Cloud.

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